



## 1. Description

### 1.1. Project

|                 |                     |
|-----------------|---------------------|
| Project Name    | Actuator Test Bench |
| Board Name      | NUCLEO-F446RE       |
| Generated with: | STM32CubeMX 6.13.0  |
| Date            | 09/12/2026          |

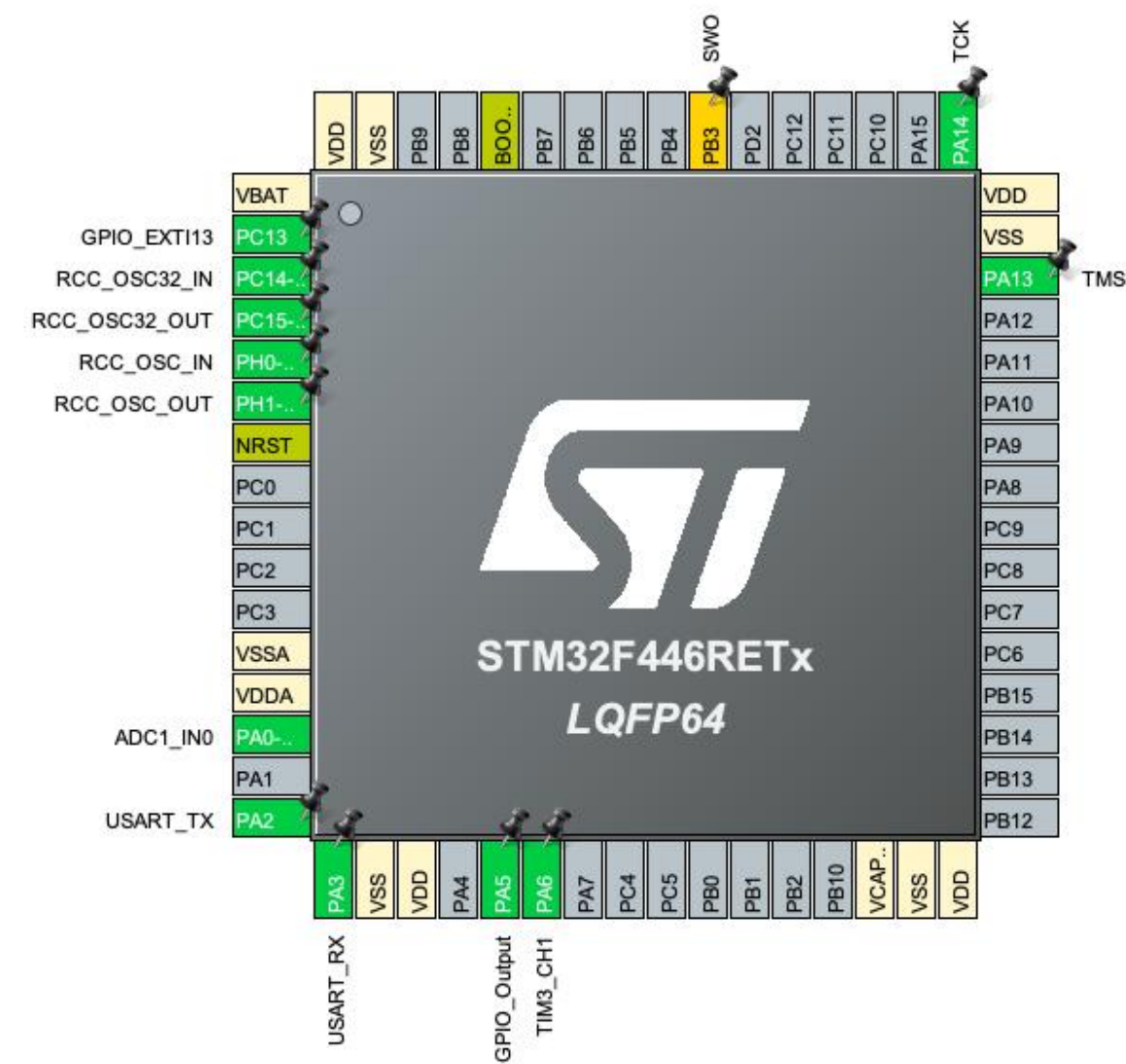
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32F4       |
| MCU Line       | STM32F446     |
| MCU name       | STM32F446RETx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | Arm Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



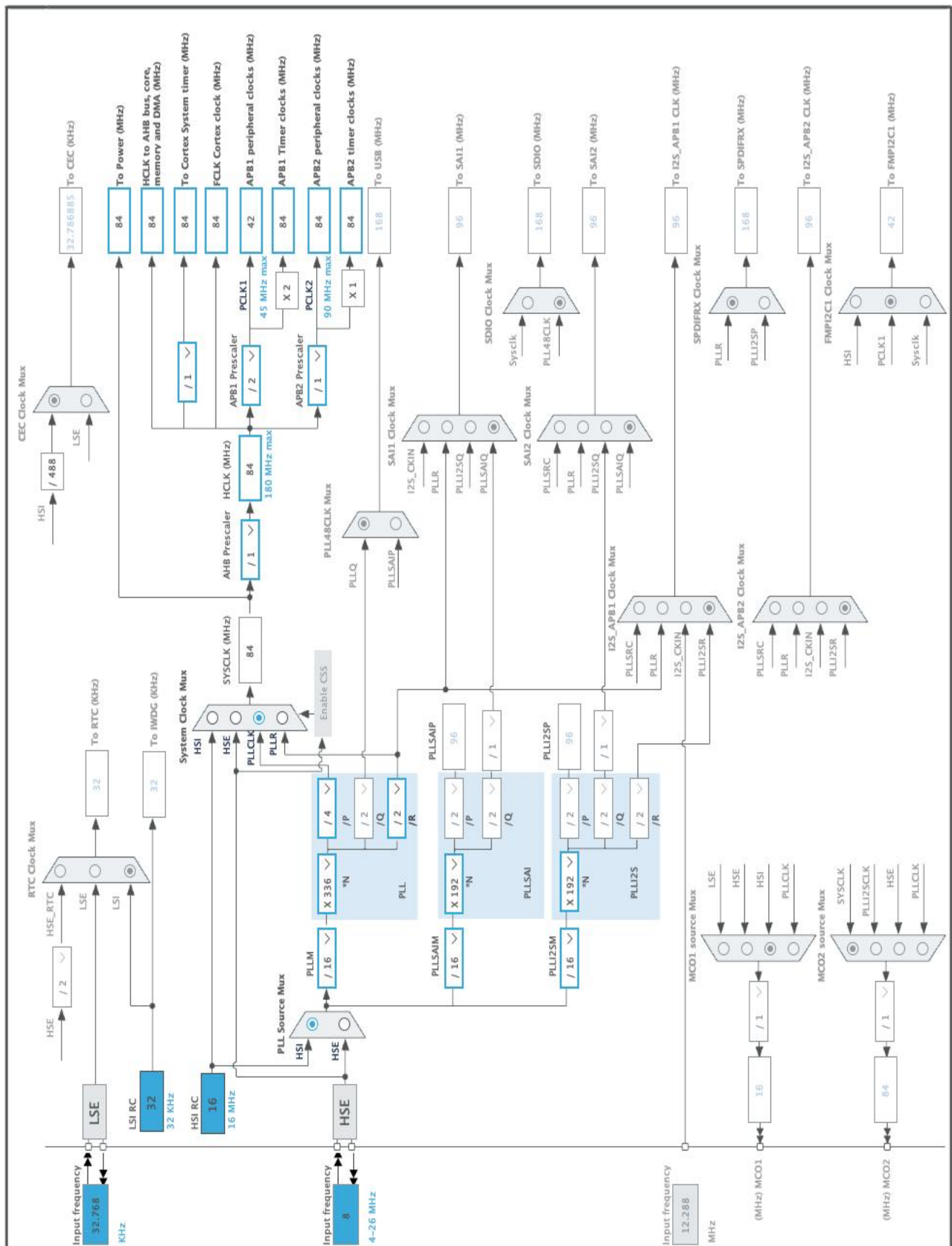
### 3. Pins Configuration

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label    |
|----------------------|---------------------------------------|----------|--------------------------|----------|
| 1                    | VBAT                                  | Power    |                          |          |
| 2                    | PC13                                  | I/O      | GPIO_EXTI13              |          |
| 3                    | PC14-OSC32_IN                         | I/O      | RCC_OSC32_IN             |          |
| 4                    | PC15-OSC32_OUT                        | I/O      | RCC_OSC32_OUT            |          |
| 5                    | PH0-OSC_IN                            | I/O      | RCC_OSC_IN               |          |
| 6                    | PH1-OSC_OUT                           | I/O      | RCC_OSC_OUT              |          |
| 7                    | NRST                                  | Reset    |                          |          |
| 12                   | VSSA                                  | Power    |                          |          |
| 13                   | VDDA                                  | Power    |                          |          |
| 14                   | PA0-WKUP                              | I/O      | ADC1_IN0                 |          |
| 16                   | PA2                                   | I/O      | USART2_TX                | USART_TX |
| 17                   | PA3                                   | I/O      | USART2_RX                | USART_RX |
| 18                   | VSS                                   | Power    |                          |          |
| 19                   | VDD                                   | Power    |                          |          |
| 21                   | PA5 *                                 | I/O      | GPIO_Output              |          |
| 22                   | PA6                                   | I/O      | TIM3_CH1                 |          |
| 30                   | VCAP_1                                | Power    |                          |          |
| 31                   | VSS                                   | Power    |                          |          |
| 32                   | VDD                                   | Power    |                          |          |
| 46                   | PA13                                  | I/O      | SYS_JTMS-SWDIO           | TMS      |
| 47                   | VSS                                   | Power    |                          |          |
| 48                   | VDD                                   | Power    |                          |          |
| 49                   | PA14                                  | I/O      | SYS_JTCK-SWCLK           | TCK      |
| 55                   | PB3 **                                | I/O      | SYS_JTDO-SWO             | SWO      |
| 60                   | BOOT0                                 | Boot     |                          |          |
| 63                   | VSS                                   | Power    |                          |          |
| 64                   | VDD                                   | Power    |                          |          |

\* The pin is affected with an I/O function

\*\* The pin is affected with a peripheral function but no peripheral mode is activated

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32F4       |
| Line      | STM32F446     |
| MCU       | STM32F446RETx |
| Datasheet | DS10693 Rev6  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.3 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

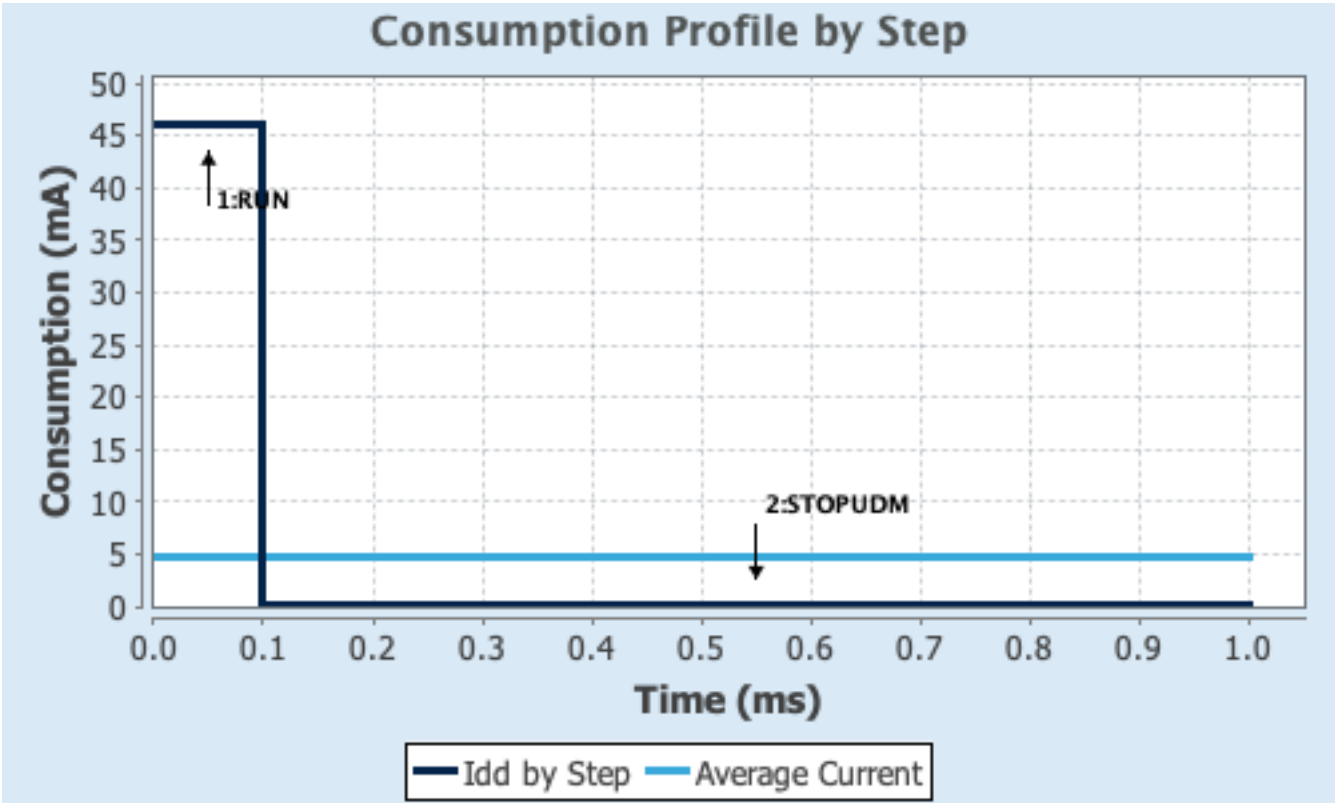
#### 1.4. Sequence

|                               |                                  |                           |
|-------------------------------|----------------------------------|---------------------------|
| <b>Step</b>                   | Step1                            | Step2                     |
| <b>Mode</b>                   | RUN                              | STOP_UDM (Under Drive)    |
| <b>Vdd</b>                    | 3.3                              | 3.3                       |
| <b>Voltage Source</b>         | Battery                          | Battery                   |
| <b>Range</b>                  | Scale1-High                      | No Scale                  |
| <b>Fetch Type</b>             | RAM/FLASH/REGON/ART/P<br>REFETCH | n/a                       |
| <b>CPU Frequency</b>          | 180 MHz                          | 0 Hz                      |
| <b>Clock Configuration</b>    | HSE PLL                          | Regulator_LP Flash-PwrDwn |
| <b>Clock Source Frequency</b> | 4 MHz                            | 0 Hz                      |
| <b>Peripherals</b>            |                                  |                           |
| <b>Additional Cons.</b>       | 0 mA                             | 0 mA                      |
| <b>Average Current</b>        | 46 mA                            | 55 $\mu$ A                |
| <b>Duration</b>               | 0.1 ms                           | 0.9 ms                    |
| <b>DMIPS</b>                  | 225.0                            | 0.0                       |
| <b>Ta Max</b>                 | 98.02                            | 104.99                    |
| <b>Category</b>               | In DS Table                      | In DS Table               |

#### 1.5. Results

|               |         |                 |             |
|---------------|---------|-----------------|-------------|
| Sequence Time | 1 ms    | Average Current | 4.65 mA     |
| Battery Life  | 1 month | Average DMIPS   | 225.0 DMIPS |

#### 1.6. Chart



## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                                             |
|-----------------------------------|-------------------------------------------------------------------|
| Project Name                      | Actuator_Test_Bench                                               |
| Project Folder                    | /Users/jiahaoli/STM32CubeIDE/workspace_1.17.0/Actuator_Test_Bench |
| Toolchain / IDE                   | STM32CubeIDE                                                      |
| Firmware Package Name and Version | STM32Cube FW_F4 V1.28.3                                           |
| Application Structure             | Advanced                                                          |
| Generate Under Root               | Yes                                                               |
| Do not generate the main()        | No                                                                |
| Minimum Heap Size                 | 0x200                                                             |
| Minimum Stack Size                | 0x400                                                             |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name       | Peripheral Instance Name |
|------|---------------------|--------------------------|
| 1    | SystemClock_Config  | RCC                      |
| 2    | MX_GPIO_Init        | GPIO                     |
| 3    | MX_USART2_UART_Init | USART2                   |
| 4    | MX_TIM3_Init        | TIM3                     |
| 5    | MX_TIM2_Init        | TIM2                     |
| 6    | MX_ADC1_Init        | ADC1                     |

## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

mode: IN0

#### 3.1.1. Parameter Settings:

##### ADCs\_Common\_Settings:

Mode Independent mode

##### ADC\_Settings:

Clock Prescaler PCLK2 divided by 4

Resolution 12 bits (15 ADC Clock cycles)

Data Alignment Right alignment

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

DMA Continuous Requests Disabled

End Of Conversion Selection EOC flag at the end of single channel conversion

##### ADC\_Regular\_ConversionMode:

Number Of Conversion 1

External Trigger Conversion Source Regular Conversion launched by software

External Trigger Conversion Edge None

Rank 1

Channel Channel 0

Sampling Time **84 Cycles \***

##### ADC\_Injected\_ConversionMode:

Number Of Conversions 0

##### WatchDog:

Enable Analog WatchDog Mode false

### 3.2. RCC

**High Speed Clock (HSE): Crystal/Ceramic Resonator**

**Low Speed Clock (LSE) : Crystal/Ceramic Resonator**

#### 3.2.1. Parameter Settings:

##### System Parameters:

VDD voltage (V) 3.3

Instruction Cache Enabled

Prefetch Buffer Enabled

Data Cache Enabled

Flash Latency(WS) 2 WS (3 CPU cycle)

**RCC Parameters:**

HSI Calibration Value 16  
TIM Prescaler Selection Disabled  
HSE Startup Timeout Value (ms) 100  
LSE Startup Timeout Value (ms) 5000

**Power Parameters:**

Power Regulator Voltage Scale Power Regulator Voltage Scale 3  
Power Over Drive Disabled

### 3.3. SYS

**Debug: Serial Wire**

**Timebase Source: SysTick**

### 3.4. TIM2

**Clock Source : Internal Clock**

#### 3.4.1. Parameter Settings:

**Counter Settings:**

Prescaler (PSC - 16 bits value) **8399 \***  
Counter Mode Up  
Counter Period (AutoReload Register - 32 bits value ) **99 \***  
Internal Clock Division (CKD) No Division  
auto-reload preload Disable

**Trigger Output (TRGO) Parameters:**

Master/Slave Mode (MSM bit) Disable (Trigger input effect not delayed)  
Trigger Event Selection Reset (UG bit from TIMx\_EGR)

### 3.5. TIM3

**Channel1: PWM Generation CH1**

#### 3.5.1. Parameter Settings:

**Counter Settings:**

Prescaler (PSC - 16 bits value) **83 \***  
Counter Mode Up  
Counter Period (AutoReload Register - 16 bits value ) **999 \***

|                                          |                                            |
|------------------------------------------|--------------------------------------------|
| Internal Clock Division (CKD)            | No Division                                |
| auto-reload preload                      | Disable                                    |
| <b>Trigger Output (TRGO) Parameters:</b> |                                            |
| Master/Slave Mode (MSM bit)              | Disable (Trigger input effect not delayed) |
| Trigger Event Selection                  | Reset (UG bit from TIMx_EGR)               |
| <b>PWM Generation Channel 1:</b>         |                                            |
| Mode                                     | PWM mode 1                                 |
| Pulse (16 bits value)                    | <b>500 *</b>                               |
| Output compare preload                   | Enable                                     |
| Fast Mode                                | Disable                                    |
| CH Polarity                              | High                                       |

## 3.6. USART2

### Mode: Asynchronous

#### 3.6.1. Parameter Settings:

|                             |                           |
|-----------------------------|---------------------------|
| <b>Basic Parameters:</b>    |                           |
| Baud Rate                   | 115200                    |
| Word Length                 | 8 Bits (including Parity) |
| Parity                      | None                      |
| Stop Bits                   | 1                         |
| <b>Advanced Parameters:</b> |                           |
| Data Direction              | Receive and Transmit      |
| Over Sampling               | 16 Samples                |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP                    | Pin            | Signal         | GPIO mode                                                          | GPIO pull/up pull down      | Max Speed   | User Label |
|-----------------------|----------------|----------------|--------------------------------------------------------------------|-----------------------------|-------------|------------|
| ADC1                  | PA0-WKUP       | ADC1_IN0       | Analog mode                                                        | No pull-up and no pull-down | n/a         |            |
| RCC                   | PC14-OSC32_IN  | RCC_OSC32_IN   | n/a                                                                | n/a                         | n/a         |            |
|                       | PC15-OSC32_OUT | RCC_OSC32_OUT  | n/a                                                                | n/a                         | n/a         |            |
|                       | PH0-OSC_IN     | RCC_OSC_IN     | n/a                                                                | n/a                         | n/a         |            |
|                       | PH1-OSC_OUT    | RCC_OSC_OUT    | n/a                                                                | n/a                         | n/a         |            |
| SYS                   | PA13           | SYS_JTMS-SWDIO | n/a                                                                | n/a                         | n/a         | TMS        |
|                       | PA14           | SYS_JTCK-SWCLK | n/a                                                                | n/a                         | n/a         | TCK        |
| TIM3                  | PA6            | TIM3_CH1       | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low         |            |
| USART2                | PA2            | USART2_TX      | Alternate Function Push Pull                                       | No pull-up and no pull-down | Very High * | USART_TX   |
|                       | PA3            | USART2_RX      | Alternate Function Push Pull                                       | No pull-up and no pull-down | Very High * | USART_RX   |
| Single Mapped Signals | PB3            | SYS_JTDO-SWO   | n/a                                                                | n/a                         | n/a         | SWO        |
| GPIO                  | PC13           | GPIO_EXTI13    | <b>External Interrupt Mode with Falling edge trigger detection</b> | No pull-up and no pull-down | n/a         |            |
|                       | PA5            | GPIO_Output    | Output Push Pull                                                   | No pull-up and no pull-down | Low         |            |

### 4.2. DMA configuration

nothing configured in DMA service

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Pre-fetch fault, memory access fault    | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 0                    | 0           |
| TIM2 global interrupt                   | true   | 0                    | 0           |
| USART2 global interrupt                 | true   | 0                    | 0           |
| EXTI line[15:10] interrupts             | true   | 0                    | 0           |
| PVD interrupt through EXTI line 16      | unused |                      |             |
| Flash global interrupt                  | unused |                      |             |
| RCC global interrupt                    | unused |                      |             |
| ADC1, ADC2 and ADC3 interrupts          | unused |                      |             |
| TIM3 global interrupt                   | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Pre-fetch fault, memory access fault    | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |
| TIM2 global interrupt                   | false                             | true                 | true             |
| USART2 global interrupt                 | false                             | true                 | true             |
| EXTI line[15:10] interrupts             | false                             | true                 | true             |

**\* User modified value**

## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

Middleware

| System Core | Analog | Timers | Connectivity | Multimedia | Computing |
|-------------|--------|--------|--------------|------------|-----------|
| DMA         | ADC1 ✓ | TIM2 ✓ | USART2 ✓     |            |           |
| GPIO ⚠      |        | TIM3 ✓ |              |            |           |
| NVIC ✓      |        |        |              |            |           |
| RCC ✓       |        |        |              |            |           |
| SYS ✓       |        |        |              |            |           |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32f446_bsdl.zip">https://www.st.com/resource/en/bsdl_model/stm32f446_bsdl.zip</a>                                                                                                                                                                                                                           |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32f446_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32f446_ibis.zip</a>                                                                                                                                                                                                                           |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32f4-svd.zip">https://www.st.com/resource/en/svd/stm32f4-svd.zip</a>                                                                                                                                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                                                                       |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                                                                       |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                                                                       |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                                                                                         |
| Product Certifications  | <a href="https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf">https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf</a>                                                                                                                                                                               |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf</a>                           |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                                                         |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an2945-stm8s-and-stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an2945-stm8s-and-stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf</a>                               |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf</a> |

- Application Notes [https://www.st.com/resource/en/application\\_note/an3364-migration-and-compatibility-guidelines-for-stm32-microcontroller-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3364-migration-and-compatibility-guidelines-for-stm32-microcontroller-applications-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an3997-audio-playback-and-recording-using-the-stm32f4discovery-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3997-audio-playback-and-recording-using-the-stm32f4discovery-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an3998-pdm-audio-software-decoding-on-stm32-microcontrollers-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3998-pdm-audio-software-decoding-on-stm32-microcontrollers-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4488-getting-started-with-stm32f4xxxx-mcu-hardware-development-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4488-getting-started-with-stm32f4xxxx-mcu-hardware-development-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4640-peripherals-interconnections-on-stm32f4057xx-stm32f4157xx-stm32f42xxx-stm32f43xxx-stm32f446xx-and-stm32f469479xx-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4640-peripherals-interconnections-on-stm32f4057xx-stm32f4157xx-stm32f42xxx-stm32f43xxx-stm32f446xx-and-stm32f469479xx-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4658-migration-of-applications-from-stm32f429439-lines-to-stm32f446-line-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4658-migration-of-applications-from-stm32f429439-lines-to-stm32f446-line-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4739-stm32cube-firmware-examples-for-stm32f4-series-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4739-stm32cube-firmware-examples-for-stm32f4-series-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf)
- Application Notes [https://www.st.com/resource/en/application\\_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf)
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